A1

H1

Antigenic ridge

Rank−biserial correlation

−1.0 −0.5 0.0 0.5 1.0

Amino acid substitution

	Temperature	Relative humidity	ΔTemperature	Δ Relative humidity
R133Q	−0.05	0.08	−0.01	0.13
R137Q	−0.02	0.03	0.11	0.05
V140A	0.05	−0.31	−0.19	−0.34
L141P	0.04	0.02	0.05	0.14
L156S	−0.16	−0.01	−0.03	0.01
L141F	−0.15	−0.03	0.23	−0.04
R133L	0.11	0.02	−0.40	0.13
R137L	−0.23	0.39	−0.17	0.27

0 10 20 30 40

A2

Epitope

Amino acid substitution

	Temperature	Relative humidity	ΔTemperature	Δ Relative humidity
E159G	−0.04	0.11	0.03	−0.11
I160N	−0.06	−0.00	−0.05	0.20
G206E	0.21	−0.23	0.11	0.15
R133Q	−0.05	0.08	−0.01	0.13
R137Q	−0.02	0.03	0.11	0.05
D209N	0.04	−0.01	0.05	−0.04
R124K	0.07	0.21	−0.02	−0.07
T132I	0.19	−0.25	0.06	0.02
G174E	0.02	0.16	−0.14	−0.01
M171I	0.07	0.09	−0.00	0.14
I172T	0.13	−0.14	−0.05	0.05
V126I	−0.08	0.05	−0.09	−0.02
T162I	−0.11	0.04	−0.28	0.02
T166I	−0.24	−0.11	−0.05	−0.11
V140A	0.05	−0.31	−0.19	−0.34
K158R	0.07	0.12	0.06	−0.06
S164N	−0.00	−0.10	0.08	−0.04
I160T	−0.12	−0.09	−0.14	0.13
L170S	0.24	0.13	0.14	0.17
L141P	0.04	0.02	0.05	0.14

0 20 40 60

Count of events across the phylogeny

B1

H3

FDR p < 0.05 n.s.

Antigenic ridge

	Temperature	Relative humidity	ΔTemperature	Δ Relative humidity
P189S	0.14	−0.20	0.15	0.03
G155R	0.04	−0.03	0.01	−0.14
P156S	0.04	−0.02	0.19	−0.11
K193E	0.38	0.20	0.09	0.31
P189T	0.16	−0.04	−0.20	−0.19

0 10 20 30

B2

Epitope

	Temperature	Relative humidity	ΔTemperature	Δ Relative humidity
H135Q	−0.10	−0.15	−0.05	−0.03
L132F	0.14	−0.01	0.12	−0.19
P189S	0.14	−0.20	0.15	0.03
Y152H	0.16	−0.03	0.07	−0.08
S273N	0.08	0.05	−0.10	0.07
I177T	0.23	−0.20	0.04	−0.06
A175T	−0.03	−0.05	0.18	0.07
A265T	−0.06	0.05	−0.07	−0.29
P150S	−0.06	0.06	0.20	−0.08
S102N	0.13	0.05	0.20	0.08
G155R	0.04	−0.03	0.01	−0.14
S137P	0.08	0.06	0.08	−0.05
D130N	−0.16	−0.05	−0.04	−0.06
D82N	0.22	−0.11	−0.09	−0.12
I261V	−0.13	0.12	−0.31	0.15
M157I	0.30	−0.18	0.21	0.02
G187R	0.11	−0.02	−0.06	0.22
Y188H	0.08	−0.14	0.07	−0.15
F57L	0.07	0.15	0.09	0.04
F122L	0.16	0.14	−0.13	0.00

0 20 40 60

Count of events across the phylogeny